Statistical Machine Learning (BE4M33SSU)
Hidden Markov Models

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- Markov models on sequences: inference & parameter learning
- Hidden Markov Models (HMMs)

Structured Hidden States

Models discussed so far: mainly classifiers predicting a categorical (class) variable $y \in \mathcal{Y}$

Often in applications: the hidden state y is a structured variable.

Here: the hidden state y is a **sequence** of categorical variables.

Application examples:

- language modelling (next token prediction),
- text recognition (printed, handwritten),
- token classification (named entity recognition, POS tagging),
- speech recognition (single word recognition, continuous speech recognition, translation),
- text-to-speech,
- bioinformatics (gene prediction),
- finance (predict stock price),
- cybersecurity (anomaly detection),
- robot self-localization.

Markov Models and Hidden Markov Models on chains:
a class of generative probabilistic models for sequences of features and sequences of categorical variables.

Markov Models: Preliminaries

Let $\boldsymbol{s} = (s_1, s_2, \ldots, s_n)$ denote a sequence of length n with elements from a finite set K.

Each element corresponds to a *discrete time step*.

Any joint probability distribution on K^n can be written as

$$p(s_1, s_2, \ldots, s_n) = p(s_1)\, p(s_2 \,|\, s_1)\, p(s_3 \,|\, s_2, s_1) \cdot \ldots \cdot p(s_n \,|\, s_1, \ldots, s_{n-1})$$

To see that, use the definition of conditional probability recursively:

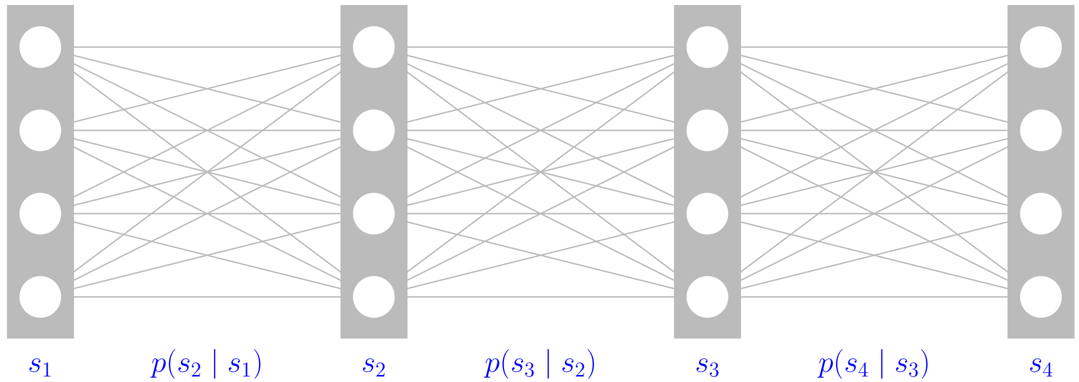
$$\begin{aligned} p(s_1, s_2, \ldots, s_n) &= p(s_n \,|\, s_1, \ldots, s_{n-1})\, p(s_1, s_2, \ldots, s_{n-1}) = \\ &= p(s_n \,|\, s_1, \ldots, s_{n-1})\, p(s_{n-1} \,|\, s_1, s_2, \ldots, s_{n-2})\, p(s_1, s_2, \ldots, s_{n-2}) = \ldots \end{aligned}$$

Markov Models: Definition

Definition 1. A joint p.d. on K^n is a *first-order* Markov model if

$$p(\boldsymbol{s}) = p(s_1)\, p(s_2 \,|\, s_1)\, p(s_3 \,|\, s_2) \cdot \ldots \cdot p(s_n \,|\, s_{n-1}) = p(s_1) \prod_{t=2}^{n} p(s_t \,|\, s_{t-1})$$

holds for any $\boldsymbol{s} = (s_1, s_2, \ldots, s_n)$.



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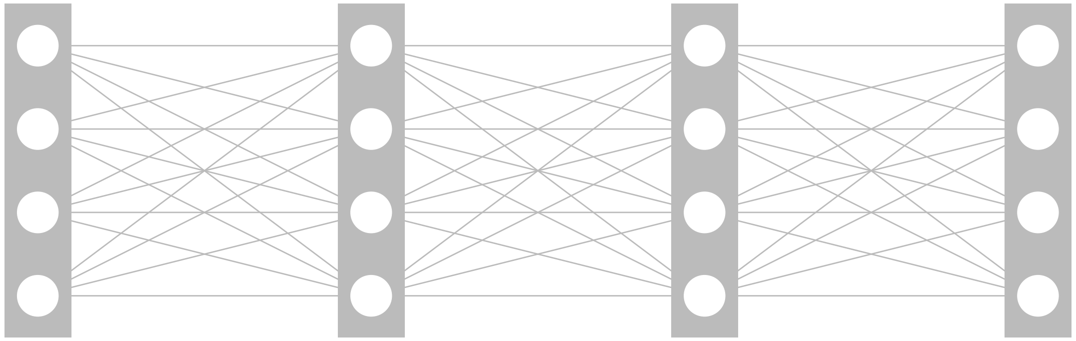

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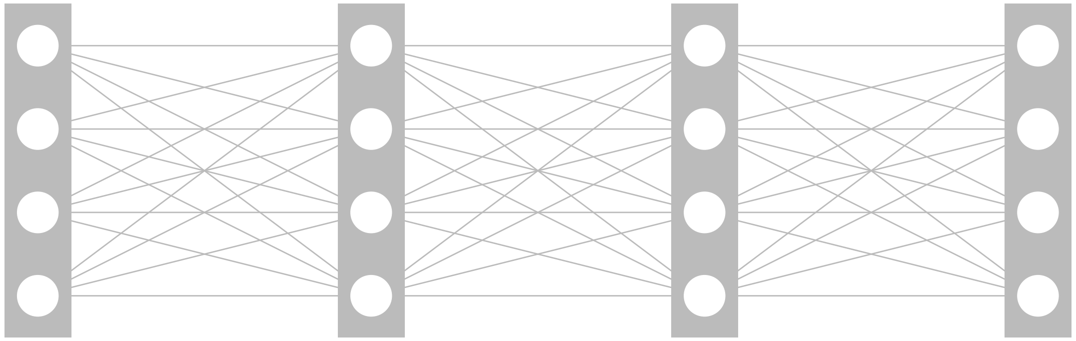

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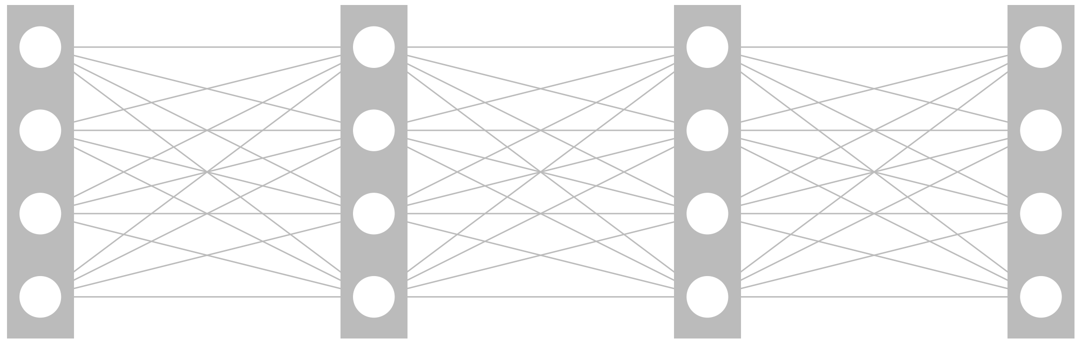

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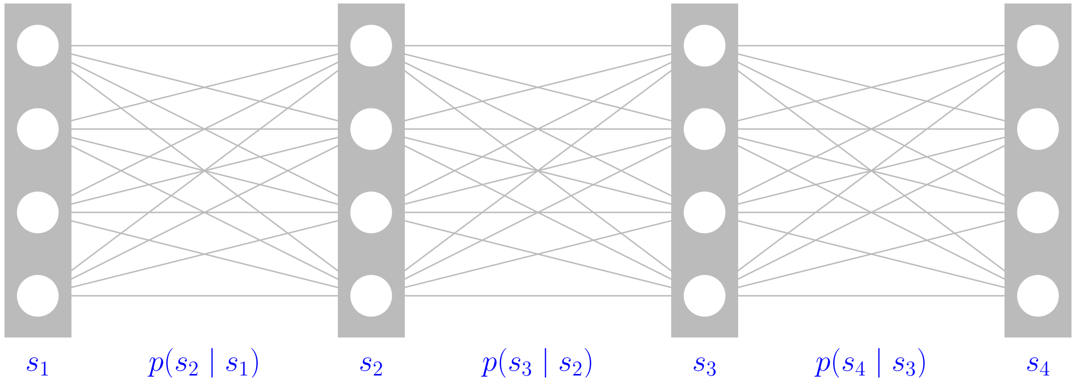

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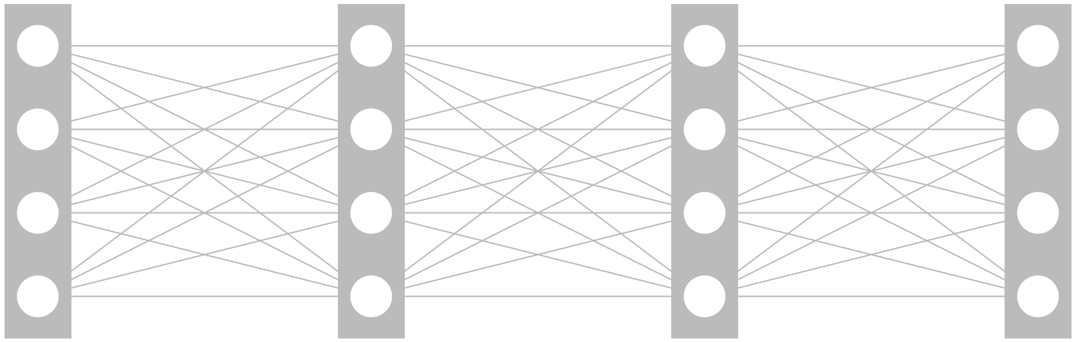

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Question 3: How many values do we need to fix the *transition probabilities* $p(s_t \,|\, s_{t-1})$?
$K \times K$ (*stochastic matrix*: columns sum to 1) $\Rightarrow K \times (K-1)$ parameters is enough.

Markov Models: Matrix Representation

The elements of a state distribution vector $\boldsymbol{\pi_t}$ for time step t are defined as:

$$(\boldsymbol{\pi}_t)_j = p(s_t = j).$$

Similarly, a transition matrix P is defined as:

$$P_{ij}(t) = p(s_t = i \,|\, s_{t-1} = j),$$

where $\sum_i P_{ij}(t) = 1$, i.e., the P is a *stochastic matrix*.

A Markov model is called homogeneous if the transition probabilities P do not depend on time. In this case, the *state distribution* at time step t can be easily written as:

$$\boldsymbol{\pi}_t = P^{t-1}\boldsymbol{\pi}_1,$$

where $\boldsymbol{\pi}_1$ is the *initial state (starting) distribution*.

Markov Models: Random Walker Example

Example 1 (Random walk on a graph)**.**

- Let (V, E) be a directed graph. A random walk in (V, E) is described by a sequence $\boldsymbol{s} = (s_1, \dots, s_t, \dots)$ of visited nodes, i.e., $s_t \in V$.
- The walker starts in node $i \in V$ with probability $p(s_1 = i)$.
- The edges of the graph are weighted by $w \colon E \to \mathbb{R}_+$, s.t.

$$\sum_{j\colon (i,j)\in E} w_{ij} = 1 \quad \forall i \in V$$

- In the current position $s_t = i$, the walker randomly chooses an outgoing edge with probability given by the weights and moves along this edge, i.e.

$$p(s_{t+1} = j \,|\, s_t = i) = \begin{cases} w_{ij} & \text{if } (i,j) \in E \\ 0 & \text{otherwise} \end{cases}$$

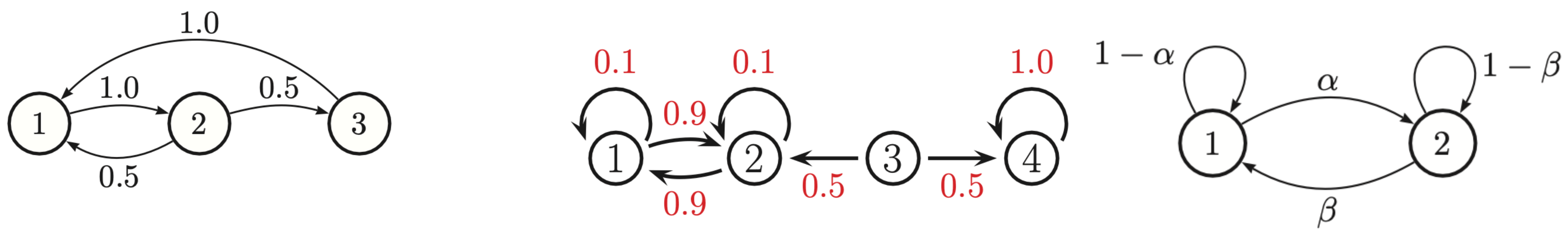
Questions: How does the distribution $p(s_t)$ behave? Does it converge to some fix-point distribution $\boldsymbol{\pi}$ for $t \to \infty$?

Return Times and Limiting Distributions

- A homogeneous Markov model is irreducible if each state l can be reached starting from any state k with non-zero probability (after some number of transitions).
- A state k has return time τ if it can be reached with non-zero probability after τ transitions when starting from itself.
- A state $k \in K$ is a-periodic if the greatest common divisor of its *return times* is 1.

Example 2. Markov chains:

- **left**: *irreducible*, *a-periodic* chain: $d(1) = d(2) = gcd(2, 3, 4, 5, 6, \ldots) = 1$, $d(3) = gcd(3, 5, 6, \ldots) = 1$.
- **middle**: *reducible* chain: starting in 4 (*absorbing state*) we end up with $\boldsymbol{\pi} = (0, 0, 0, 1)$; starting in 1 or 2 we get $\boldsymbol{\pi} = (\frac{1}{2}, \frac{1}{2}, 0, 0)$; starting in 3 leads to either of the two *stationary distributions*.
- **right**: *irreducible* chain for $\alpha, \beta = 0.9$: we end up with $\boldsymbol{\pi} = (\frac{1}{2}, \frac{1}{2})$ by symmetry. For $\alpha, \beta = 1$ it depends where you start.



Murphy, Machine Learning: a Probabilistic Perspective, 2012

Return Times and Limiting Distributions, contd.

Theorem 1. *Let P be the transition probability matrix of an* ***irreducible*** *homogeneous Markov model with all states* ***a-periodic****. Then there exists a unique* ***stationary distribution*** $\boldsymbol{\pi}^*$ *s.t.* $P\boldsymbol{\pi}^* = \boldsymbol{\pi}^*$. *Moreover, it is a* ***limiting distribution****, i.e.*

$$\lim_{t\to\infty} P^t \boldsymbol{\pi}_1 = \boldsymbol{\pi}^*$$

for arbitrary starting distributions $\boldsymbol{\pi}_1$.

Remark 2. The two assumptions for the limiting distribution:

- the **irreducibility** means that the state transition diagram must be a singly-connected component,
- the **a-periodicity** means no oscillations.

Remark 3. A special case of the above is a regular chain for which: $P^n_{ij} > 0$ for some n and all i and j:

- Hence, after n steps, we can end up in any state, no matter where we started.
- *Irreducibility* and a *self-loop* for all states is all we need.

Google PageRank

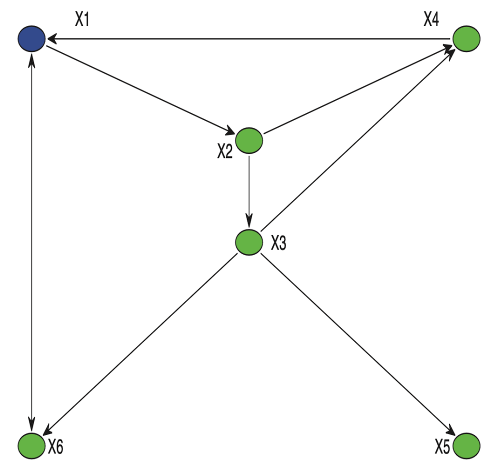
- One of the methods used to rank the retrieved documents (big in 1997).
- Idea: good pages are referenced more often; a random surfer spends more time on highly reachable pages.
- It is the *Random walk*:
  - P – transition matrix based on hyperlinks between pages,
  - π_i – page i authority (high for being linked by many other pages),
  - authority is passed to the linked pages:

$$\pi_i = \sum_j P_{ij}\,\pi_j.$$

  - Corresponds to the Markov model *stationary distribution*.
- We need the P to be *irreducible* having all states *a-periodic*:
  - Real link adjacency matrix G is sparse as not all pages are linked to all other pages.
  - We must set P, so even non-existing links (in G) get at least a very small probability, so we end up with a **regular** chain.

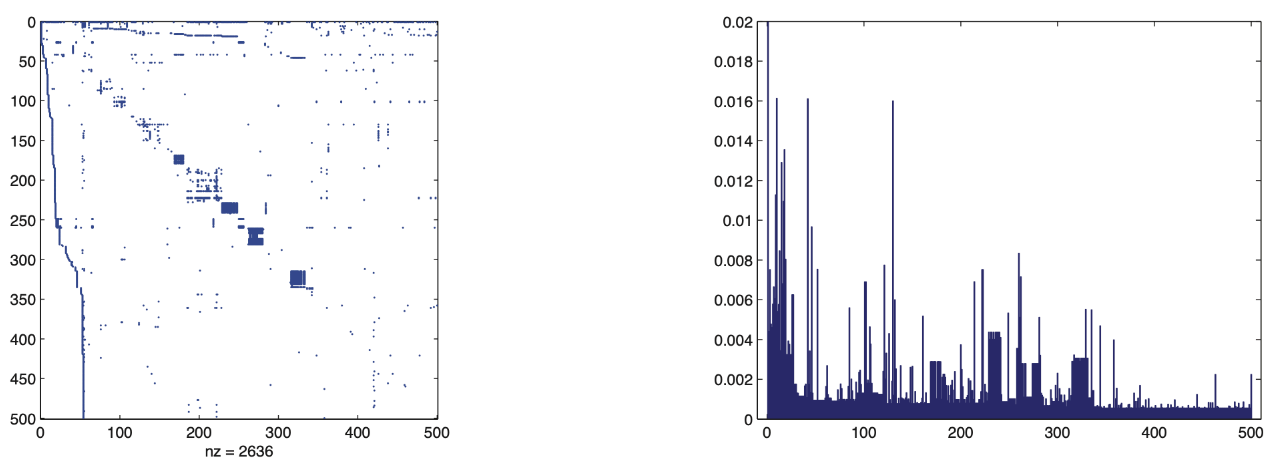
Google PageRank Example

Simple example



Converges to $\boldsymbol{\pi} = (0.3209, 0.1706, 0.1065, 0.1368, 0.0643, 0.2008)$

Example for 500 pages linked from `www.harvard.edu`:
Adjacency matrix G (left), PageRank vector $\boldsymbol{\pi}$ (right).



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Algorithms: Computing the Most Probable Sequence

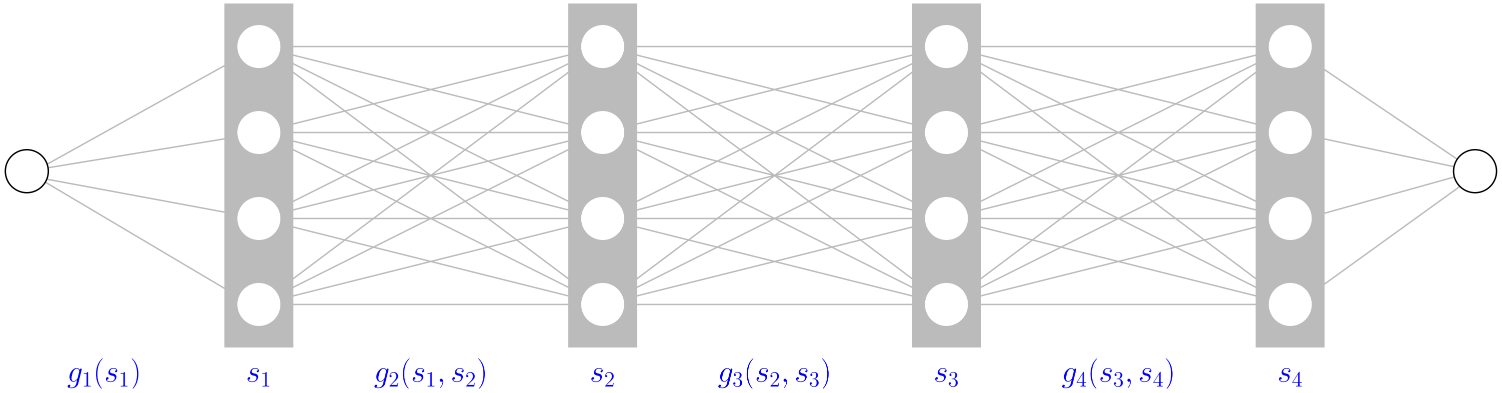
We want to compute the most probable sequence $\boldsymbol{s}^* \in \operatorname*{arg\,max}_{\boldsymbol{s} \in K^n} \Big[p(s_1) \prod_{t=2}^{n} p(s_t \,|\, s_{t-1}) \Big]$.

Take the logarithm of $p(\boldsymbol{s})$: $\boldsymbol{s}^* \in \operatorname*{arg\,max}_{\boldsymbol{s} \in K^n} \big[g_1(s_1) + \sum_{t=2}^{n} g_t(s_{t-1}, s_t) \big]$

and apply dynamic programming: Set $\phi_1(s_1) \equiv g_1(s_1)$ and compute

$$\phi_t(s_t) = \max_{s_{t-1} \in K} \big[\phi_{t-1}(s_{t-1}) + g_t(s_{t-1}, s_t) \big] \quad \forall s_t \in K.$$

Finally, find $s_n^* \in \operatorname{arg\,max}_{s_n \in K} \phi_n(s_n)$ and back-track the solution. This corresponds to searching the best path in the graph



Question: What is the run-time complexity? .

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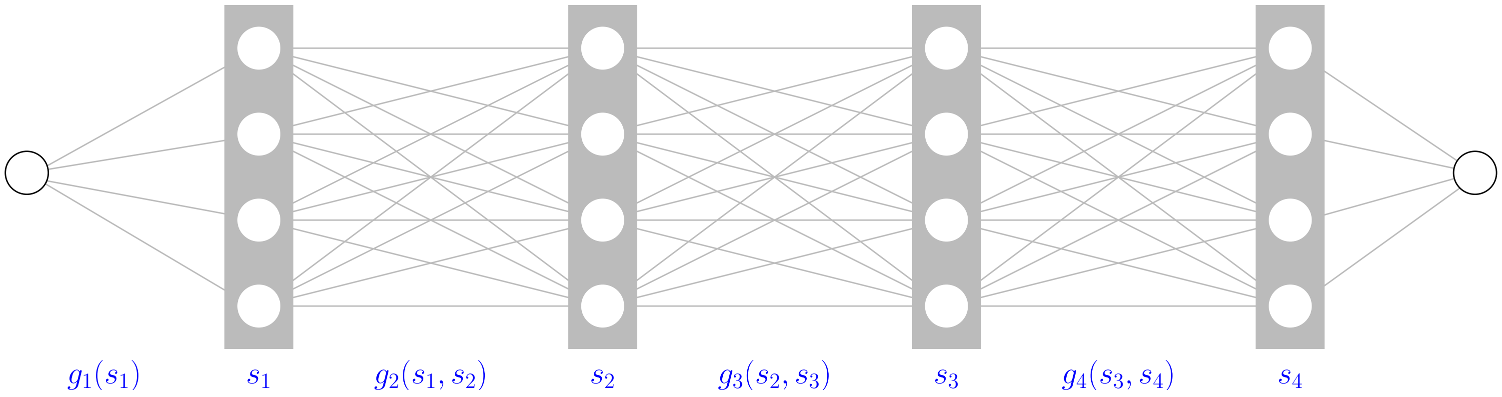
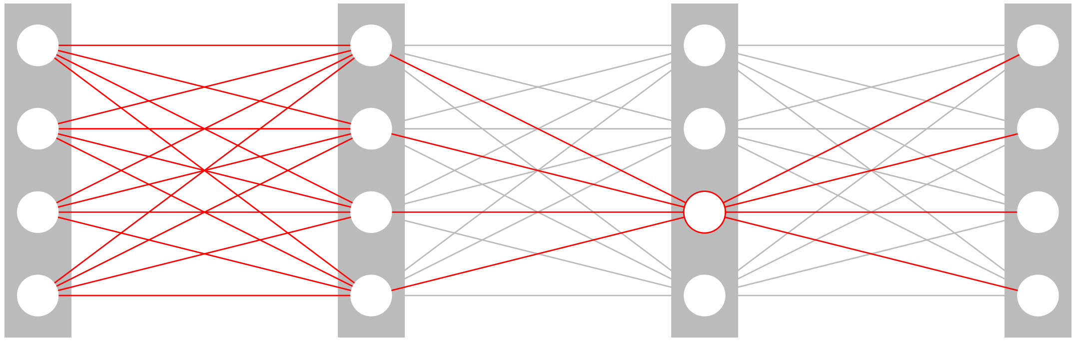


Question: What is the run-time complexity? $\mathcal{O}(nK^2)$.

Algorithms: Computing Marginal Probabilities

How to compute marginal probabilities for the sequence element s_j in position j

$$p(s_j) = \sum_{s_1 \in K} \cdots \cancel{\sum_{s_j \in K}} \cdots \sum_{s_n \in K} p(s_1) \prod_{t=2}^{n} p(s_t \,|\, s_{t-1})$$



Summation over the **trailing** variables is easily done because:

$$\sum_{s_n \in K} p(s_1) \cdots p(s_{n-1} \,|\, s_{n-2})\, p(s_n \,|\, s_{n-1}) = p(s_1) \cdots p(s_{n-1} \,|\, s_{n-2})$$

Algorithms: Computing Marginal Probabilities, contd.

The summation over the **leading** variables is done dynamically: Begin with $p(s_1)$ and compute:

$$p(s_t) = \sum_{s_{t-1} \in K} p(s_t \,|\, s_{t-1})\, p(s_{t-1}) \quad \forall s_t \in K.$$

Equivalently using matrix representation the computation reads: $\boldsymbol{\pi}_t = P(t)\boldsymbol{\pi}_{t-1}$.

Run-time complexity once again $\mathcal{O}(nK^2)$.

Remark 4.

- Notice that the preferred direction (from first to last) in the definition of a Markov model is only apparent. By computing the marginal probabilities $p(s_t)$ and by using $p(s_t \,|\, s_{t-1})p(s_{t-1}) = p(s_{t-1}, s_t) = p(s_{t-1} \,|\, s_t)p(s_t)$, we can rewrite the model in reverse order.

Algorithms: Learning a Markov Model

Suppose we are given i.i.d. training data $\mathcal{T}^m = \{\boldsymbol{s}^k \in K^n \,|\, k = 1, \ldots, m\}$ and want to estimate the parameters of the Markov model by the maximum likelihood estimate. This is very easy:

- Denote by $a(s_{t-1} = j, s_t = i)$ the number of sequences in $\mathcal{T}^m$ for which $s_{t-1} = j$ and $s_t = i$.
- The estimates for the conditional probabilities are then given by

$$p(s_t = i \,|\, s_{t-1} = j) = \frac{a(s_{t-1} = j, s_t = i)}{\sum_k a(s_{t-1} = i, s_t = k)}.$$

Algorithms: Learning a Markov Model, contd.

But is our estimate an MLE?

Let's recall the log-likelihood:

$$\frac{1}{m}\sum_{\boldsymbol{s}\in\mathcal{T}^m}\left[\log p(s_1)+\sum_{t=2}^{n}\log p(s_t\,|\,s_{t-1})\right]$$

Proof (idea):

Consider all terms in the log-likelihood that depend on the transition probability from $(t-1)\rightarrow t$ and rewrite them using transition counts $a(s_{t-1}=j, s_t=i)$

$$\frac{1}{m}\sum_{\boldsymbol{s}\in\mathcal{T}^m}\log p(s_t\,|\,s_{t-1})=\frac{1}{m}\sum_{i,j\in K}a(s_{t-1}=j,s_t=i)\log p(s_t=i\,|\,s_{t-1}=j)$$

Maximise this w.r.t. $p(s_t\,|\,s_{t-1})$ under the constraint $\sum_{s_t\in K}p(s_t\,|\,s_{t-1})=1$ using Lagrange multipliers.

Markov Models are Exponential Families

Markov models are **exponential families**. For simplicity we show this for the family of homogeneous Markov models on sequences $\boldsymbol{s} = (s_1, s_2, \ldots, s_n)$ of length n under the additional assumption that $p(s_1) = \frac{1}{K}$.

We have

$$p(\boldsymbol{s}) = \frac{1}{K} \prod_{t=2}^{n} p(s_t \,|\, s_{t-1}),$$

- sufficient statistic: $\Phi(\boldsymbol{s})$ is a $K \times K$ matrix with entries $\Phi_{kl}(\boldsymbol{s})$ counting the number of transitions from state l to state k in the sequence $\boldsymbol{s}$,
- natural parameter: H is a $K \times K$ matrix with entries $H_{kl} = \log p(s_t = k \,|\, s_{t-1} = l)$,
- base measure: $h(\boldsymbol{s}) = 1$,
- cumulant function: $\log(K)$.

We can write the probability of sequences as

$$p(\boldsymbol{s}; H) = \exp\big[\langle \Phi(\boldsymbol{s}), H \rangle - \log(K)\big]$$

Remark 5. This can be generalised for models with non-uniform $p(s_1)$ and also for general (i.e. non-homogeneous) Markov models.

Hidden Markov Models

- Let $\boldsymbol{s} = (s_1, s_2, \ldots, s_n)$ denote a sequence of hidden states from a finite set K.
- Let $\boldsymbol{x} = (x_1, x_2, \ldots, x_n)$ denote a sequence of features from some feature space $\mathcal{X}$.

Definition 2. A joint p.d. on $\mathcal{X}^n \times K^n$ is a Hidden Markov model if

(a) the prior p.d. $p(\boldsymbol{s})$ for the sequences of hidden states is a Markov model, and

(b) the conditional distribution $p(\boldsymbol{x} \,|\, \boldsymbol{s})$ for the feature sequence is independent, i.e.

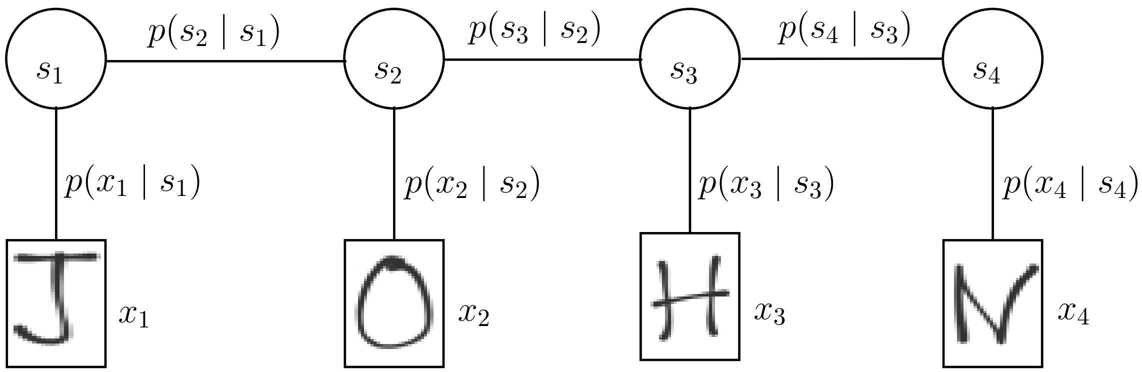
$$p(\boldsymbol{x} \,|\, \boldsymbol{s}) = \prod_{t=1}^{n} p(x_t \,|\, s_t).$$

The joint model is given by

$$p(\boldsymbol{x}, \boldsymbol{s}) = p(s_1) \prod_{t=2}^{n} p(s_t \,|\, s_{t-1}) \prod_{t=1}^{n} p(x_t \,|\, s_t).$$

Hidden Markov Model: OCR Example

Example 3 (Text recognition, OCR).

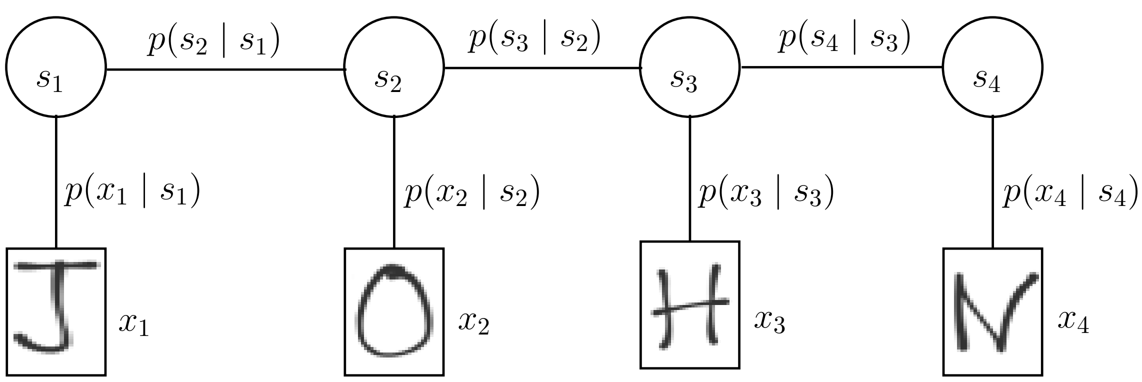


- $\boldsymbol{x} = (x_1, x_2, \ldots, x_n)$ – sequence of images with characters,
- $\boldsymbol{s} = (s_1, s_2, \ldots, s_n)$ – sequence of alphabetic characters,
- $p(s_t \,|\, s_{t-1})$ – language model (*transition probabilities*),
- $p(x_t \,|\, s_t)$ – appearance model for characters (*emission probabilities*).

Question: How to generate a sequence?

Hidden Markov: Inference Tasks

- Filtering: compute the **belief state** $p(s_t \,|\, \boldsymbol{x}_{1:t})$ *online* using **forward** algorithm. Get less noise than using just $p(s_t \,|\, x_t)$.
- Smoothing: compute $p(s_t \,|\, \boldsymbol{x}_{1:n})$ *offline* given all evidence using **forward-backward** algorithm. Even less uncertainty given future data.
- Fixed lag smoothing: $p(s_{t-l} \,|\, \boldsymbol{x}_{1:t})$, where $l > 0$ is called the lag, compromise of online and offline.
- Most probable sequence: compute $\arg\max_{\boldsymbol{s}_{1:n}} p(\boldsymbol{s}_{1:n} \,|\, \boldsymbol{x}_{1:n})$ using **Viterbi** algorithm.
- Prediction: compute $p(s_{t+h} \,|\, \boldsymbol{x}_{1:t})$:, where $h > 0$ is called the prediction **horizon**: just use the *belief state* and power of the transition matrix P^h (for a homogeneous HMM).
- Posterior samples: generate new samples $\boldsymbol{s}^s_{1:n} \sim p(\boldsymbol{s}_{1:n} \,|\, \boldsymbol{x}_{1:n})$.



Filtering: Forward Algorithm

We want to compute $\alpha_t(k) \triangleq p(s_t = k \,|\, \boldsymbol{x}_{1:t})$, $\forall k \in K$ in some sequence position t.

Let us start with one-step-ahead predictive density:

$$p(s_t = k \,|\, \boldsymbol{x}_{1:t-1}) = \sum_j \underbrace{p(s_t = k \,|\, s_{t-1} = j)}_{\Psi(j,k)} \underbrace{p(s_{t-1} = j \,|\, \boldsymbol{x}_{1:t-1})}_{\alpha_{t-1}(j)},$$

now define the update step:

$$\alpha_t(k) = p(s_t = k \,|\, \boldsymbol{x}_{1:t}) = p(s_t = k \,|\, x_t, \boldsymbol{x}_{1:t-1}) = \frac{1}{Z_t} p(x_t \,|\, s_t = k, \cancel{\boldsymbol{x}_{1:t-1}})\, p(s_t = k \,|\, \boldsymbol{x}_{1:t-1}),$$

where $Z_t = p(x_t \,|\, \boldsymbol{x}_{1:t-1}) = \sum_{k'} p(x_t \,|\, s_t = k')\, p(s_t = k' \,|\, \boldsymbol{x}_{1:t-1})$.

The matrix-vector form of the update is:

$$\boldsymbol{\alpha}_t \propto \boldsymbol{\psi}_t \odot (\boldsymbol{\Psi}^T \boldsymbol{\alpha}_{t-1}),$$

where $\boldsymbol{\psi}_t(k) \triangleq p(x_t \,|\, s_t = k)$ denotes the emission probabilities and $\odot$ is the Hadamard product (elemenwise vector multiplication). Initialize with $\boldsymbol{\alpha}_1 \propto \boldsymbol{\psi}_1 \odot \boldsymbol{\pi}_1$.

This is the forward algorithm.

Smoothing: Forward-Backward Algorithm

How to compute marginal probabilities for hidden states given the sequence of features?

$$p(s_t = k \,|\, \boldsymbol{x}) = p(s_t = k \,|\, \boldsymbol{x}_{1:t}, \boldsymbol{x}_{t+1:n}) \propto \underbrace{p(s_t = k \,|\, \boldsymbol{x}_{1:t})}_{\text{forward}} \underbrace{p(\boldsymbol{x}_{t+1:n} \,|\, s_t = k, \cancel{\boldsymbol{x}_{1:t}})}_{\text{backward}}$$

We know how to compute the *forward*, now let us look at the *backward* part, firstly define the conditional likelihood of future evidence given that the hidden state at time i:

$$\beta_t(k) \triangleq p(\boldsymbol{x}_{t+1:n} \,|\, s_t = k).$$

Let us establish the recurrence:

$$\beta_{t-1}(k) = p(\boldsymbol{x}_{t:n} \,|\, s_{t-1} = k) = \sum_j p(s_t = j, x_t, \boldsymbol{x}_{t+1:n} \,|\, s_{t-1} = k) =$$

$$= \sum_j p(\boldsymbol{x}_{t+1:n} \,|\, s_t = j, \cancel{s_{t-1} = k}, \cancel{x_t})\, p(s_t = j, x_t \,|\, s_{t-1} = k) =$$

$$= \sum_j p(\boldsymbol{x}_{t+1:n} \,|\, s_t = j)\, p(x_t \,|\, s_t = j, \cancel{s_{t-1} = k})\, p(s_t = j \,|\, s_{t-1} = k) = \sum_j \beta_t(j) \psi_t(j) \Psi(k, j).$$

The backward algorithm is then $\boldsymbol{\beta}_{t-1} \propto \boldsymbol{\Psi}(\boldsymbol{\psi}_t \odot \boldsymbol{\beta}_t)$, with $\beta_n(k) = p(\underbrace{\boldsymbol{x}_{n+1:n}}_{\emptyset} \,|\, s_n = k) = 1$.

Smoothing: Forward-Backward Algorithm, contd.

The forward-backward algorithm now reads:

$$p(s_t = k \,|\, \boldsymbol{x}) \propto \alpha_t(k)\, \beta_t(k),$$

where both $\alpha_t(k)$ and $\beta_t(k)$ are computed using dynamic programming. The result must be normalized.

The computational complexity for computing $\alpha_t(k)$ and $\beta_t(k)$ in all positions $t = 1, \ldots, n$ is thus $\mathcal{O}(nK^2)$.

Two-Slice Smoothed Marginals

We can also compute pairwise marginal probabilities $p(s_t, s_{t+1} \,|\, \boldsymbol{x})$ from the α-s and β-s. Start with:

$$\begin{aligned} p(s_t, s_{t+1}, \boldsymbol{x}) &= p(\boldsymbol{x}_{1:t} \,|\, s_t, \cancel{s_{t+1}})\, p(x_{t+1} \,|\, \cancel{s_t}, s_{t+1})\, p(\boldsymbol{x}_{t+2:n} \,|\, \cancel{s_t}, s_{t+1})\, p(s_{t+1} \,|\, s_t)\, p(s_t) \\ &= p(s_t \,|\, \boldsymbol{x}_{1:t})\, p(\boldsymbol{x}_{1:t})\, p(x_{t+1} \,|\, s_{t+1})\, p(\boldsymbol{x}_{t+2:n} \,|\, s_{t+1})\, p(s_{t+1} \,|\, s_t) \end{aligned}$$

Now we get:

$$p(s_t = i, s_{t+1} = j \,|\, \boldsymbol{x}) = \frac{p(s_t = i, s_{t+1} = j, \boldsymbol{x})}{p(\boldsymbol{x})} \propto \alpha_t(i)\, p(x_{t+1} \,|\, s_{t+1})\, p(s_{t+1} \,|\, s_t)\, \beta_{t+1}(j).$$

To summarize the computation decomposes to:

- $\alpha_t(i)$: past evidence,
- $p(s_{t+1} \,|\, s_t)$: transition to the *current* state,
- $p(x_{t+1} \,|\, s_{t+1})$: *current* observation,
- $\beta_{t+1}(j)$: future observations.

The computational complexity remains $\mathcal{O}(nK^2)$.

Most Probable Sequence: the Viterbi Algorithm

How to find the most probable sequence of hidden states given the sequence of features $\boldsymbol{x}$

$$\boldsymbol{s}^* \in \operatorname*{arg\,max}_{\boldsymbol{s}\in K^n} \; p(s_1)\prod_{t=2}^{n} p(s_t \,|\, s_{t-1}) \prod_{t=1}^{n} p(x_t \,|\, s_t)$$

Take the logarithm, with

$$g_1(s_1, x_1) = \log[p(s_1)p(x_1 \,|\, s_1)]$$
$$g_i(s_{t-1}, s_t, x_t) = \log[p(s_t \,|\, s_{t-1})p(x_t \,|\, s_t)], \; t = 2, \ldots, n$$

Solve the task

$$\boldsymbol{s}^* \in \operatorname*{arg\,max}_{\boldsymbol{s}\in K^n} \Big[g_1(s_1, x_1) + \sum_{t=2}^{n} g_t(s_{t-1}, s_t, x_t)\Big]$$

by dynamic programming as before for Markov models.

The computational complexity is again $\mathcal{O}(nK^2)$.

Viterbi vs. Smoothing

Compare the most probable sequence computer using the Viterbi algorithm:

$$\boldsymbol{s}^* \in \operatorname*{arg\,max}_{\boldsymbol{s}\in K^n} \; p(s_1)\prod_{t=2}^{n} p(s_t \,|\, s_{t-1}) \prod_{t=1}^{n} p(x_t \,|\, s_t)$$

with the smoothed marginals computed using forward-backward:

$$\hat{\boldsymbol{s}} \triangleq \left(\operatorname*{arg\,max}_{s_1\in K} p(s_1 \,|\, \boldsymbol{x}), \operatorname*{arg\,max}_{s_2\in K} p(s_2 \,|\, \boldsymbol{x}), \ldots, \operatorname*{arg\,max}_{s_n\in K} p(s_n \,|\, \boldsymbol{x}) \right).$$

They are not the same! Example with two time steps and observations $=$ hidden states:

	$x_1 = 0$	$x_1 = 1$	
$x_2 = 0$	0.04	0.3	0.34
$x_2 = 1$	0.36	0.3	0.66
	0.4	0.6	

Viterbi: $(0, 1)$ is globaly optimal, smoothed marginals $(1, 1)$ localy optimal.

Sampling Sequences from the Posterior

The task is to:

$$\boldsymbol{s}^s_{1:n} \sim p(\boldsymbol{s}_{1:n} \,|\, \boldsymbol{x}_{1:n})$$

We can follow these steps:

- Compute the *two-slice smoothed marginals* $p(s_t, s_{t+1} \,|\, \boldsymbol{x})$.
- Normalize them to get $p(s_{t+1} \,|\, s_t, \boldsymbol{x})$.
- Sample from the initial pair of states: $\boldsymbol{s}^s_{1:2} \sim p(s_1, s_2 \,|\, \boldsymbol{x})$.
- Recursively sample $s^s_{t+1} \sim p(s_{t+1} \,|\, s_t, \boldsymbol{x})$.

Note this is forwards-backwards pass and a additional sampling pass. It is possible to do it in a single pass...

Learning algorithms for HMMs

Supervised learning:

Given i.i.d. training data $\mathcal{T}^m = \{(\boldsymbol{x}^j, \boldsymbol{s}^j) \in \mathcal{X}^n \times K^n \,|\, j = 1, \ldots, m\}$, estimate the parameters of the HMM by the maximum likelihood estimator.

This is done by simple "counting" as before for Markov models.

- Denote by $a_t(s_{t-1} = j, s_t = i)$ the number of examples in $\mathcal{T}^m$ for which $s_{t-1} = j$ and $s_t = i$.
- Denote by $b_t(s_t = i, x_t = x)$ the number of examples in $\mathcal{T}^m$ for which $s_t = i$ and $x_t = x$.

The estimates for the model parameters are then given by

$$p(s_t = i \,|\, s_{t-1} = j) = \frac{a_t(s_{t-1} = j, s_t = i)}{\sum_{i'} a_t(s_{t-1} = j, s_t = i')}, \; p(x_t = x \,|\, s_t = i) = \frac{b_t(s_t = i, x_t = x)}{\sum_{x'} b_t(s_t = k, x_t = x')}$$

Remark 6. This is easy to generalise for the case that $\mathcal{X} = \mathbb{R}^m$ and $p(x_t \,|\, s_t)$ is from some parametric distribution family.

Learning algorithms for HMMs

Unsupervised learning:

Given i.i.d. training data $\mathcal{T}^m = \{\boldsymbol{x}^j \in \mathcal{X}^n \,|\, j = 1, \ldots, m\}$, estimate the parameters of the HMM by the maximum likelihood estimator.

We apply the EM-algorithm (aka Baum-Welch algorithm): Initialise the model, then iterate

E-step Use the current model estimate and compute the two-slice smoothed marginals

$$\alpha_{\boldsymbol{x}}(s_t, s_{t-1}) = p(s_t, s_{t-1} \,|\, \boldsymbol{x}) \quad s_{t-1}, s_t \in K$$

for each training example $\boldsymbol{x} \in \mathcal{T}^m$ and all positions $i = 2, \ldots, n$.

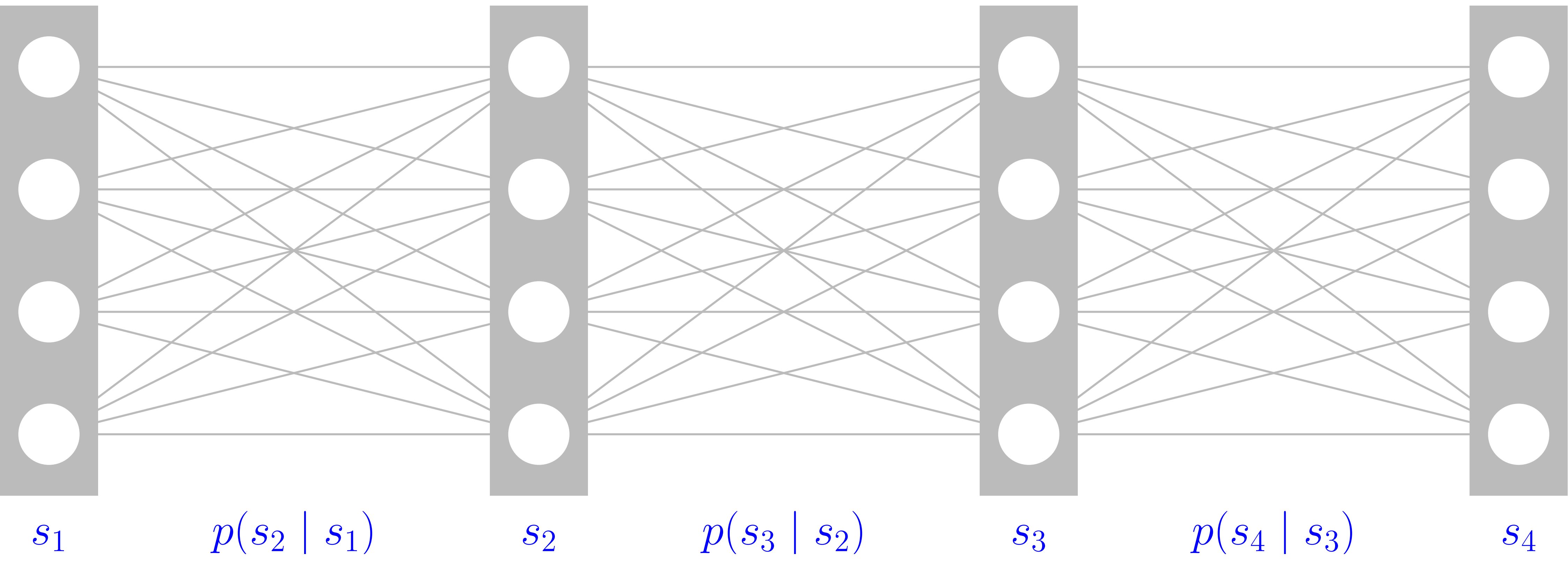
M-step Use the α-s as soft labels for computing the counts as in supervised learning

$$a_t(s_{t-1}, s_t) = \sum_{\boldsymbol{x} \in \mathcal{T}^m} \alpha_{\boldsymbol{x}}(s_t, s_{t-1}), \qquad b_t(s_t, x) = \sum_{\boldsymbol{x} \in \mathcal{T}(x_t = x)} \alpha_{\boldsymbol{x}}(s_t)$$

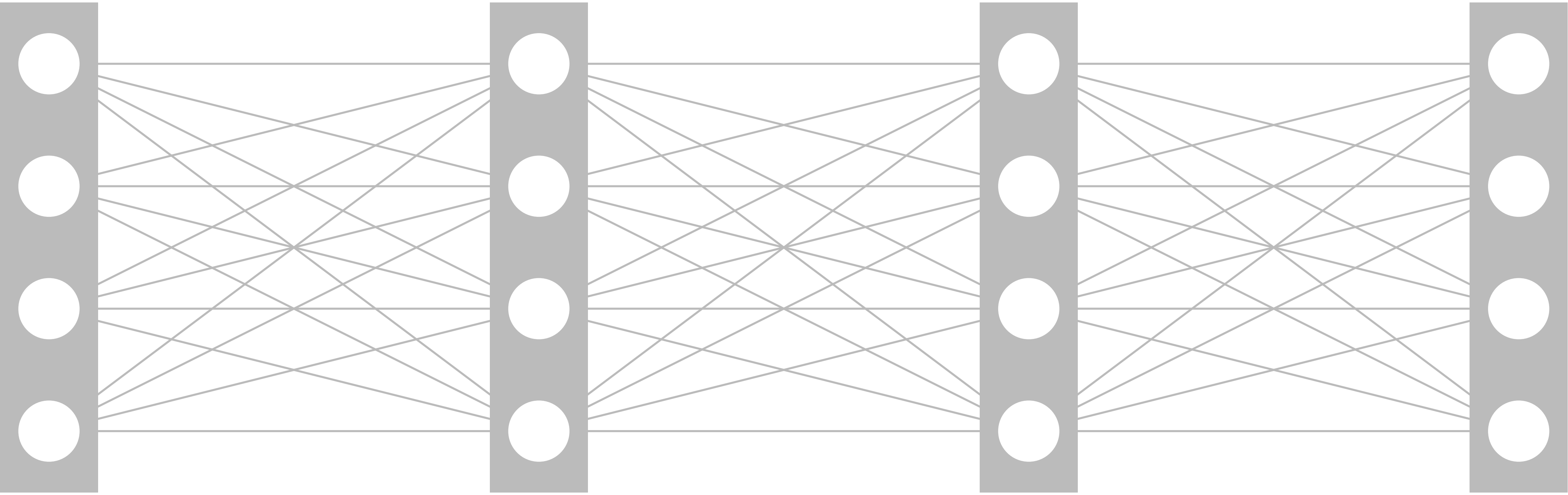
where $\mathcal{T}(x_t = x)$ is the set of training examples with $x_t = x$ and $\alpha_{\boldsymbol{x}}(s_t)$ is given by $\alpha_{\boldsymbol{x}}(s_t) = \sum_{s_{t-1}} \alpha_{\boldsymbol{x}}(s_t, s_{t-1})$.

Summary

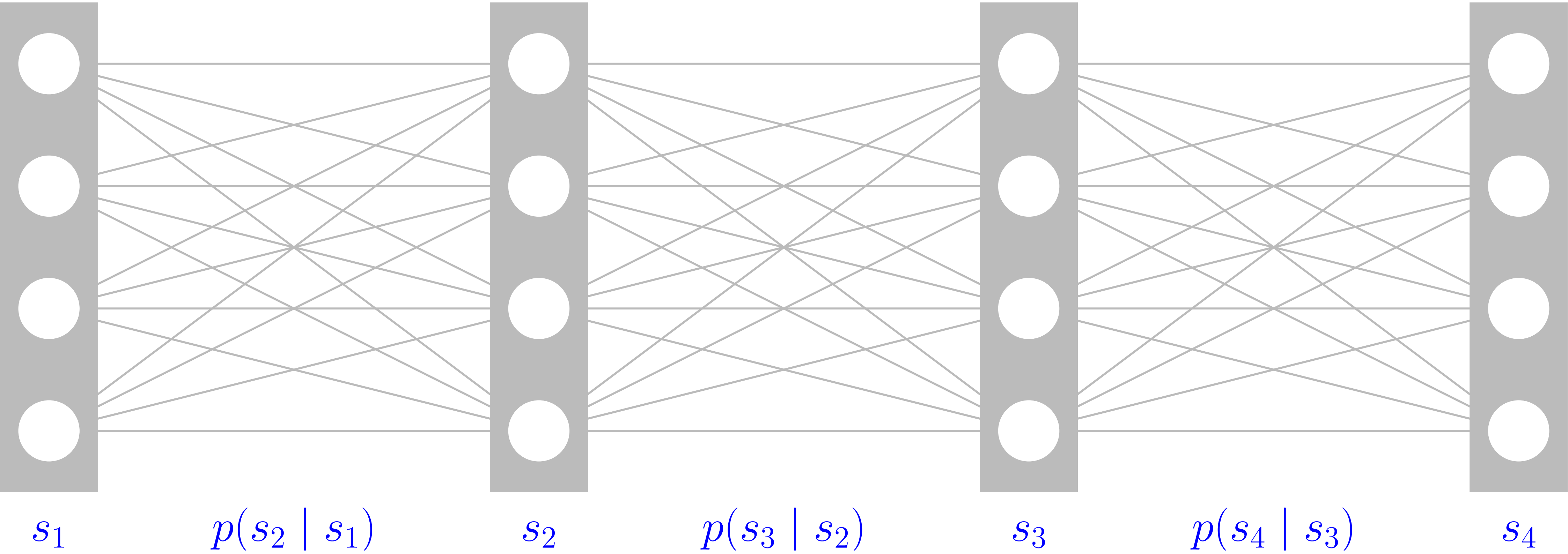
- Hidden Markov models are statistical models for pairs of processes (sequences) $(\boldsymbol{s}, \boldsymbol{x})$, where $\boldsymbol{s}$ is a sequence of hidden states and not directly observable.
- Markov models and HMMs are exponential families.
- Their parameters can be estimated by supervised leaning using either Maximum likelihood estimates or Empirical risk minimization.
- Their parameters can be estimated by unsupervised learning using the EM-algorithm.
- All important inference and learning algorithms for HMMs have time complexity linear in the length of the sequence and quadratic in the size of the hidden state space.
- What if the space of hidden states is very large, e.g. $s_t \in \mathbb{Z}^k$? We can use recurrent neural networks for modelling $p(s_t \,|\, s_{t-1})$, e.g. Gated recurrent Units (GRU) or Long short-term memory models (LSTM).



s_1
$p(s_2 \mid s_1)$
s_2
$p(s_3 \mid s_2)$
s_3
$p(s_4 \mid s_3)$
s_4



s_1 $\quad p(s_2 \mid s_1) \quad$ s_2 $\quad p(s_3 \mid s_2) \quad$ s_3 $\quad p(s_4 \mid s_3) \quad$ s_4


s_1
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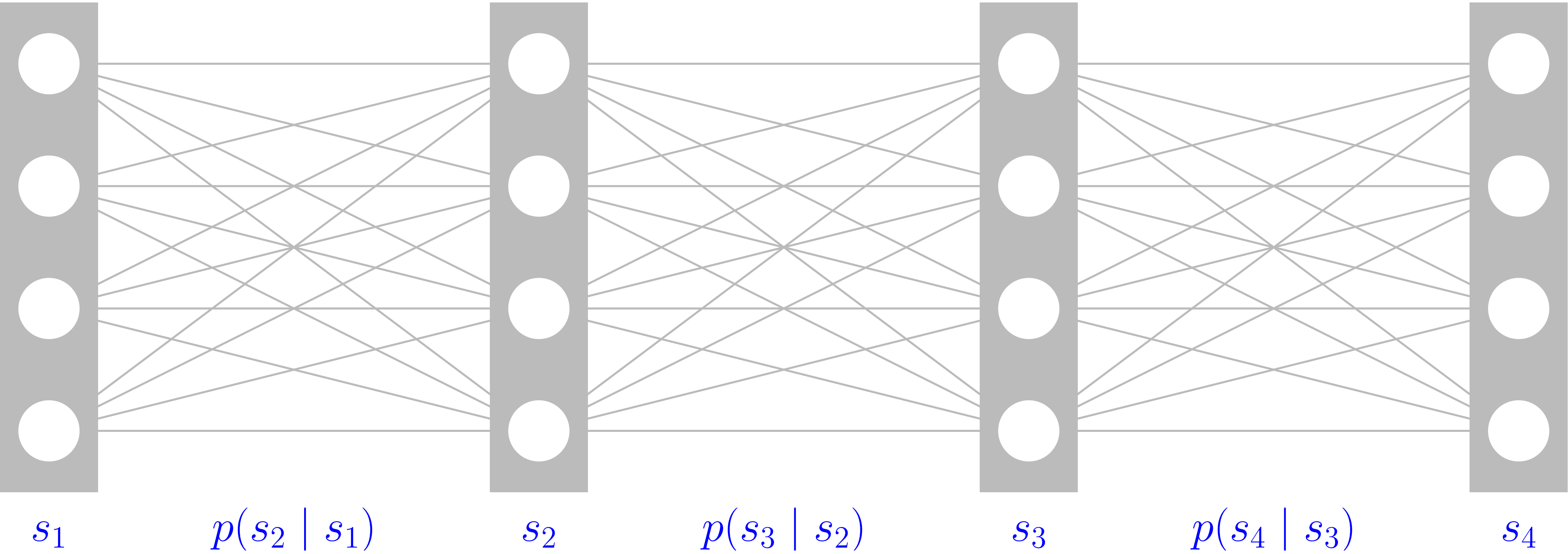
s_3
$p(s_4 \mid s_3)$
s_4


s_1
$p(s_2 \mid s_1)$
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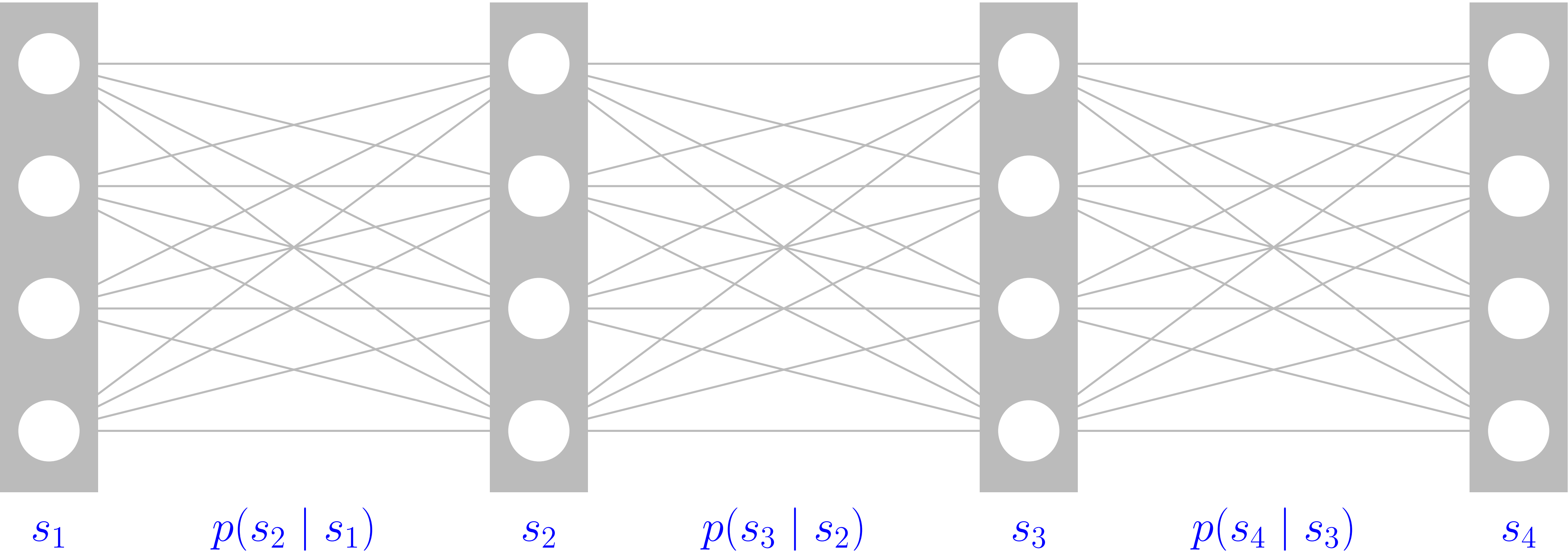
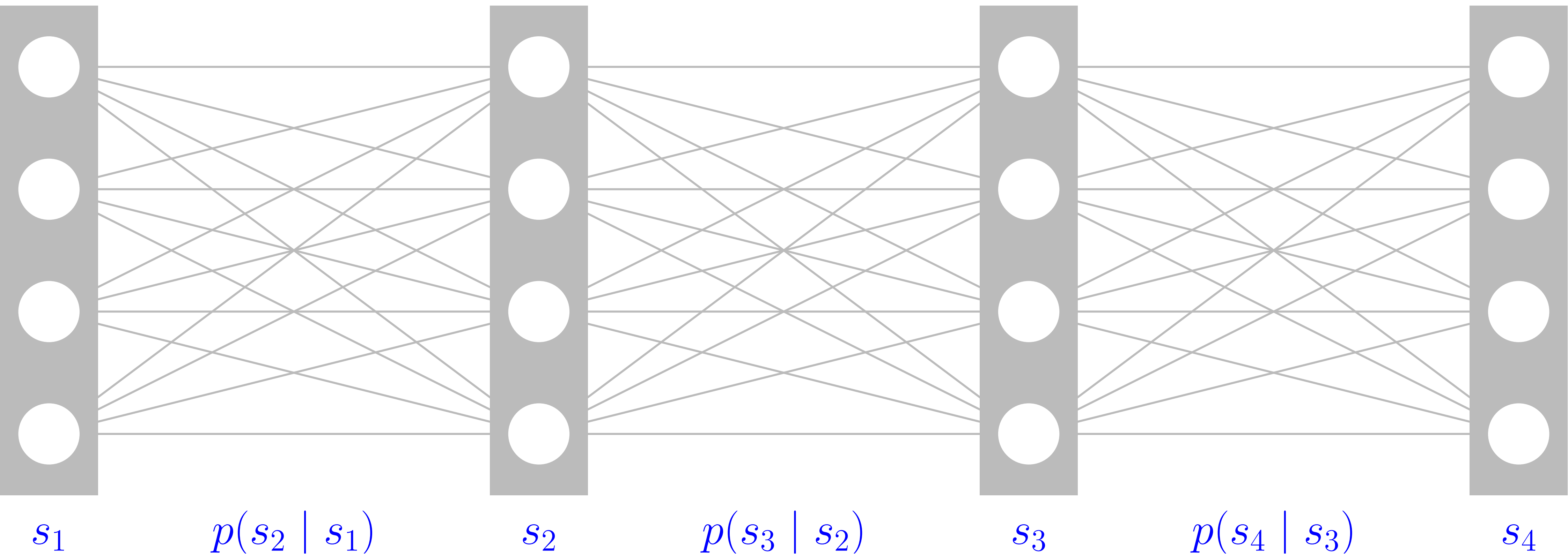
$p(s_3 \mid s_2)$
s_3
$p(s_4 \mid s_3)$
s_4


s_1
$p(s_2 \mid s_1)$
s_2
$p(s_3 \mid s_2)$
s_3
$p(s_4 \mid s_3)$
s_4


s_1
$p(s_2 \mid s_1)$
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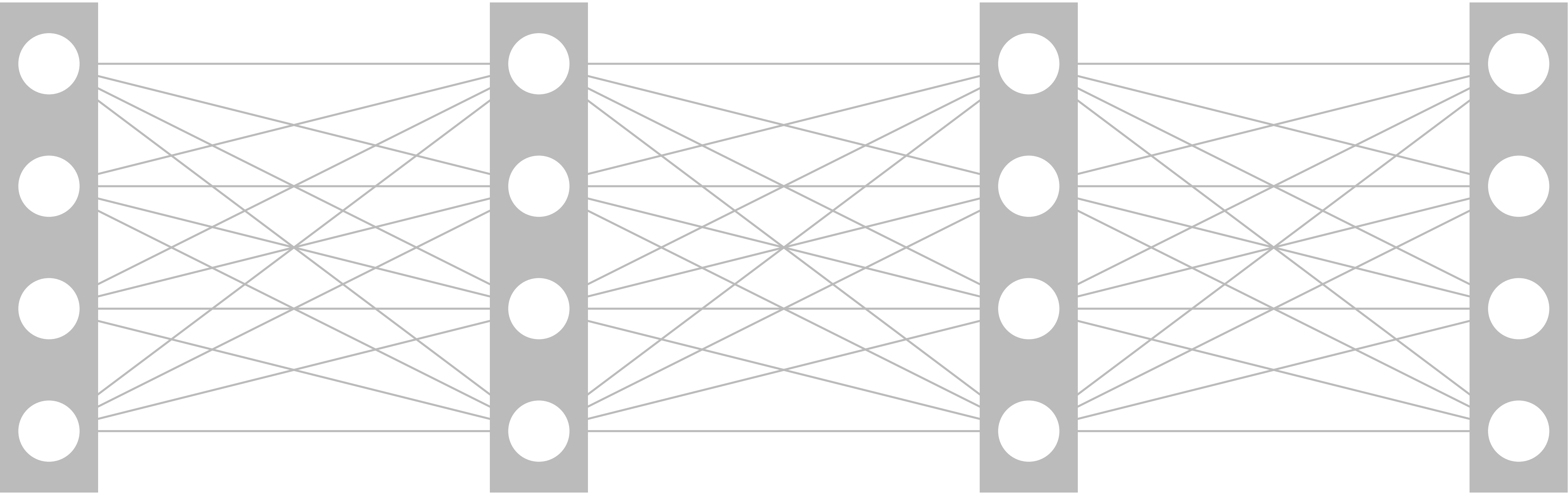
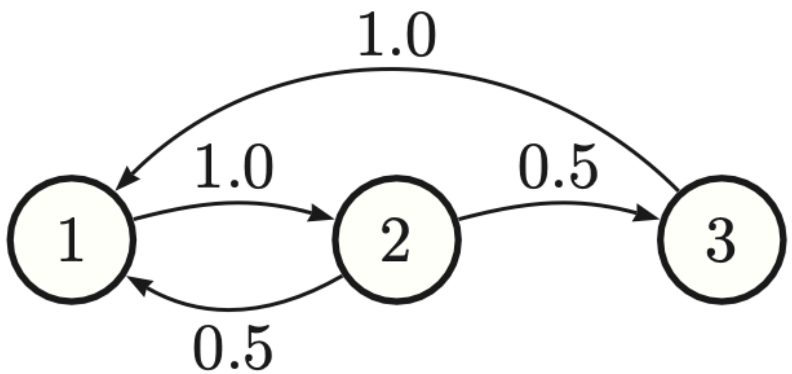
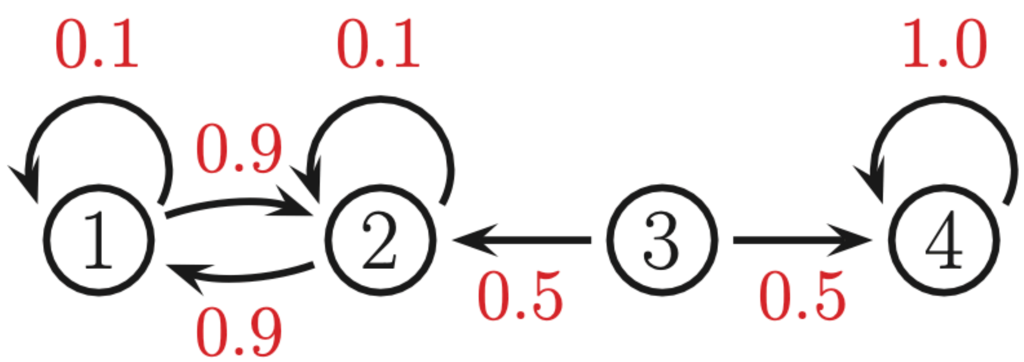
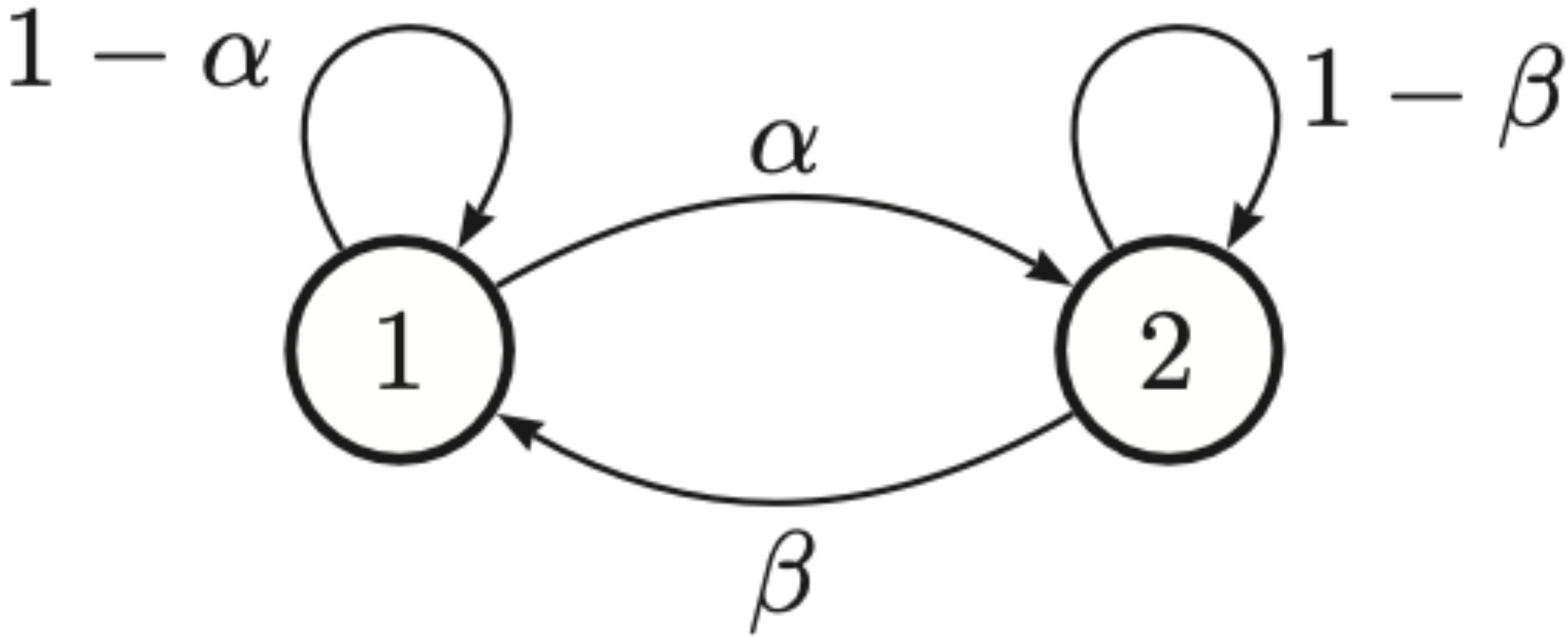
$p(s_3 \mid s_2)$
s_3
$p(s_4 \mid s_3)$
s_4

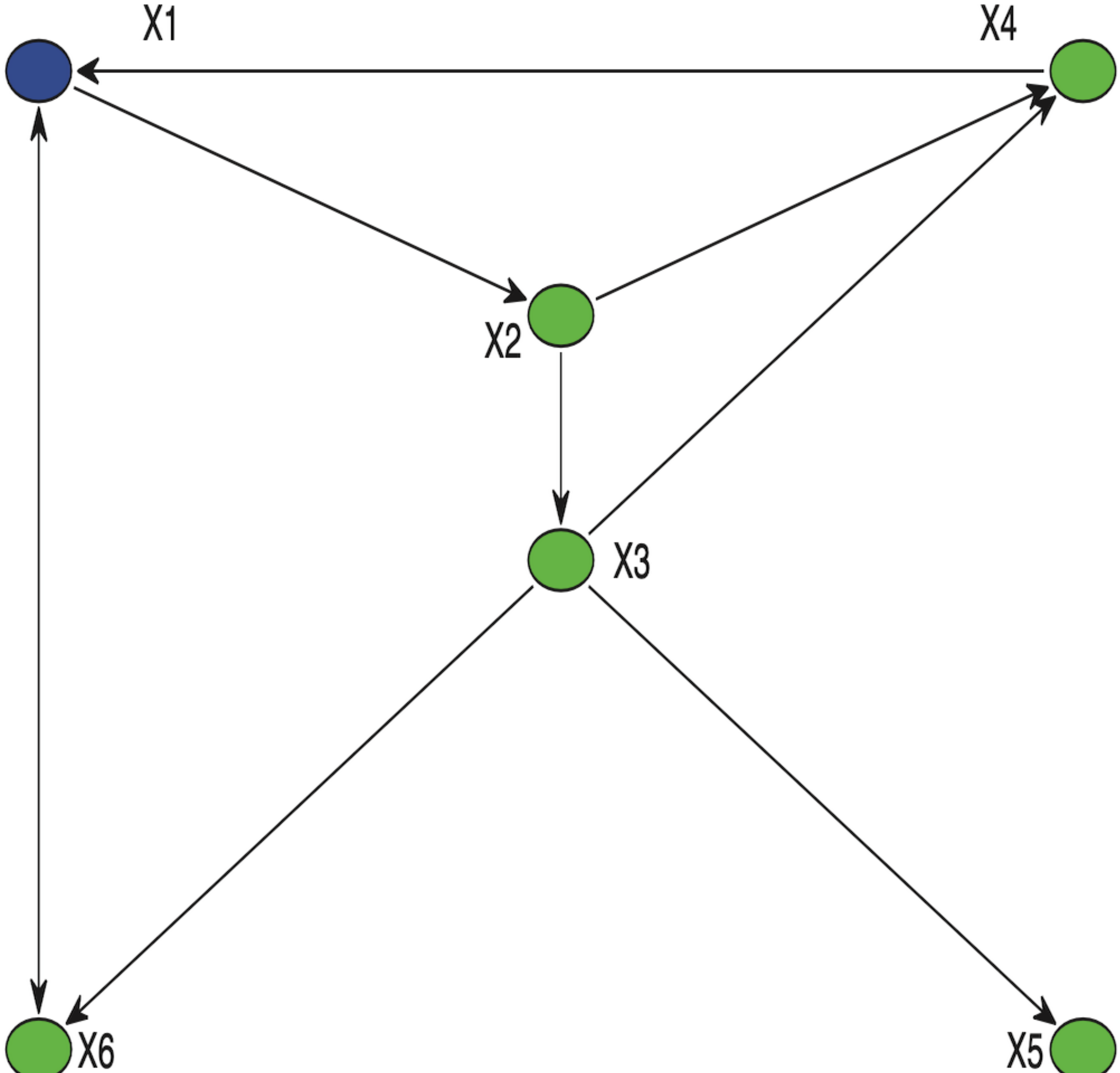


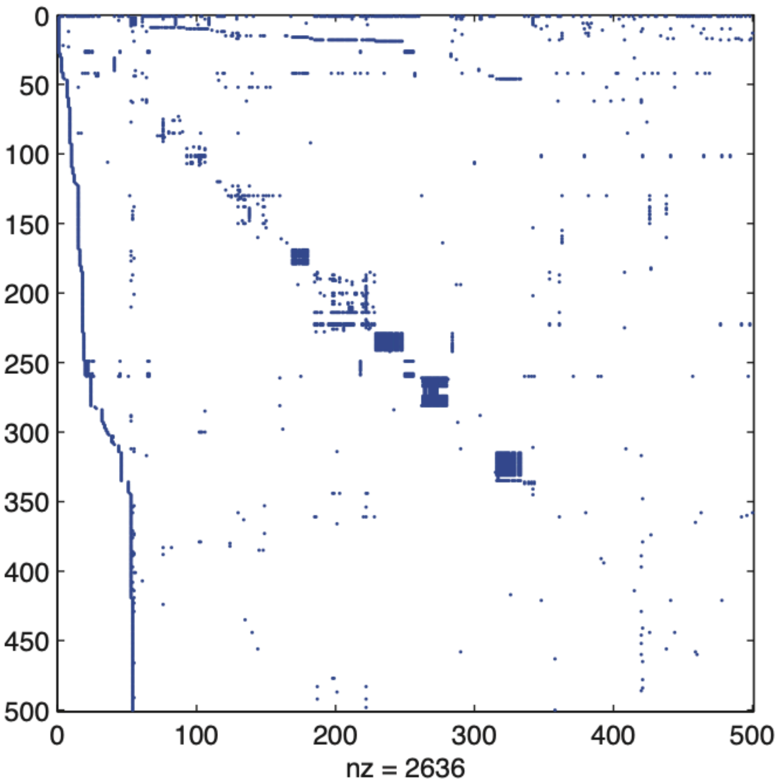
s_1 $\quad p(s_2 \mid s_1)$ $\quad s_2$ $\quad p(s_3 \mid s_2)$ $\quad s_3$ $\quad p(s_4 \mid s_3)$ $\quad s_4$


1.0
1.0
0.5
1
2
3
0.5


0.1
0.1
1.0
0.9
1
2
3
4
0.9
0.5
0.5


$1 - \alpha$
α
$1 - \beta$
1
2
β


X1
X4
X2
X3
X6
X5



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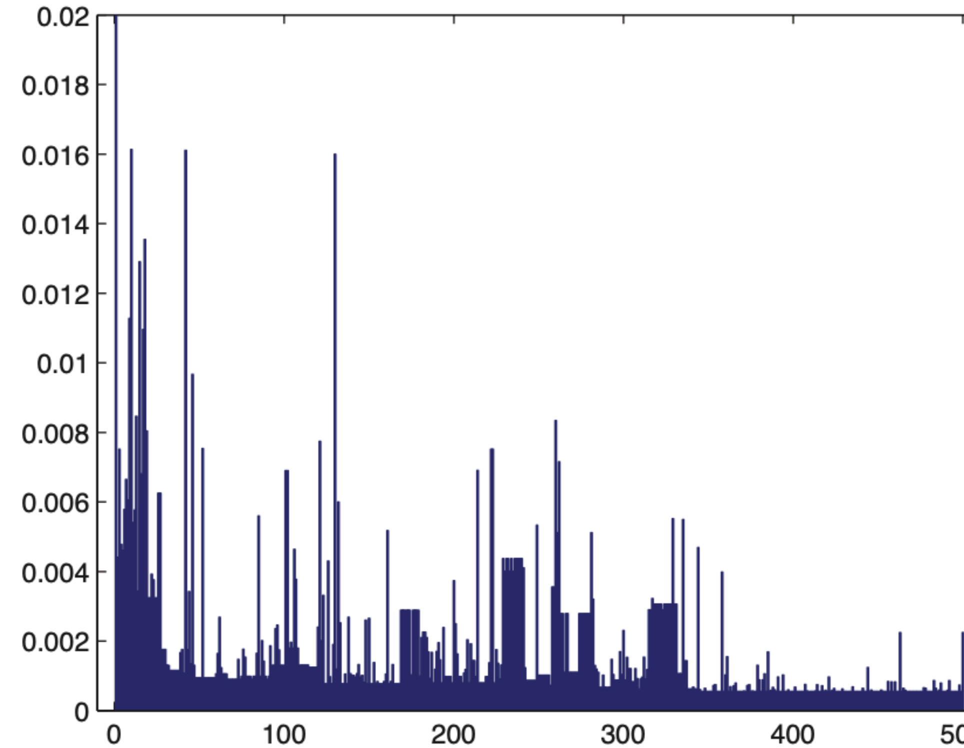
nz = 2636



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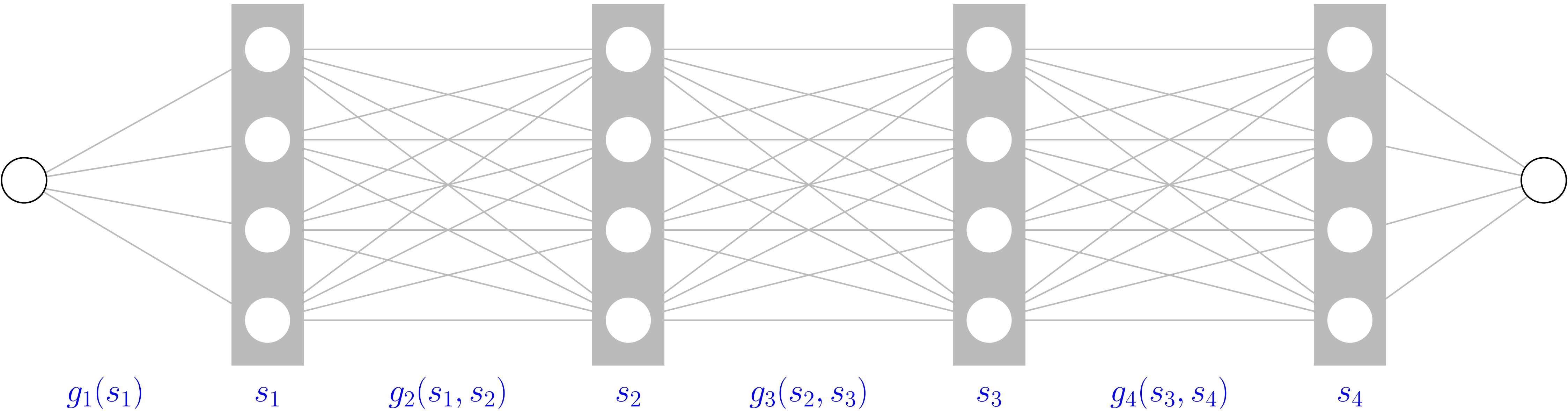

$g_1(s_1)$
s_1
$g_2(s_1, s_2)$
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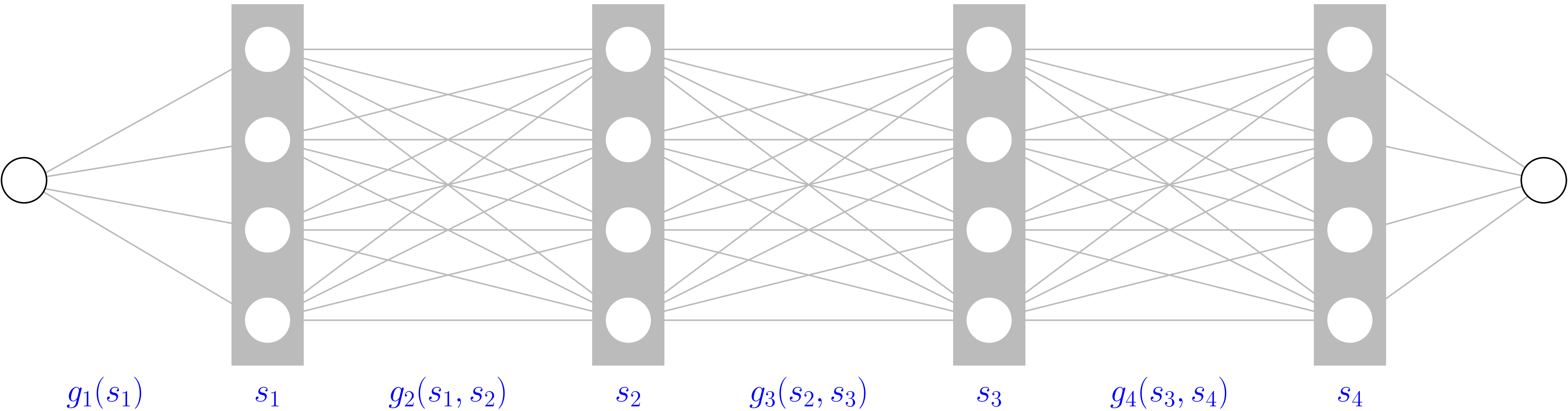
$g_3(s_2, s_3)$
s_3
$g_4(s_3, s_4)$
s_4


$g_1(s_1)$
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$g_2(s_1, s_2)$
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$g_3(s_2, s_3)$
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$g_4(s_3, s_4)$
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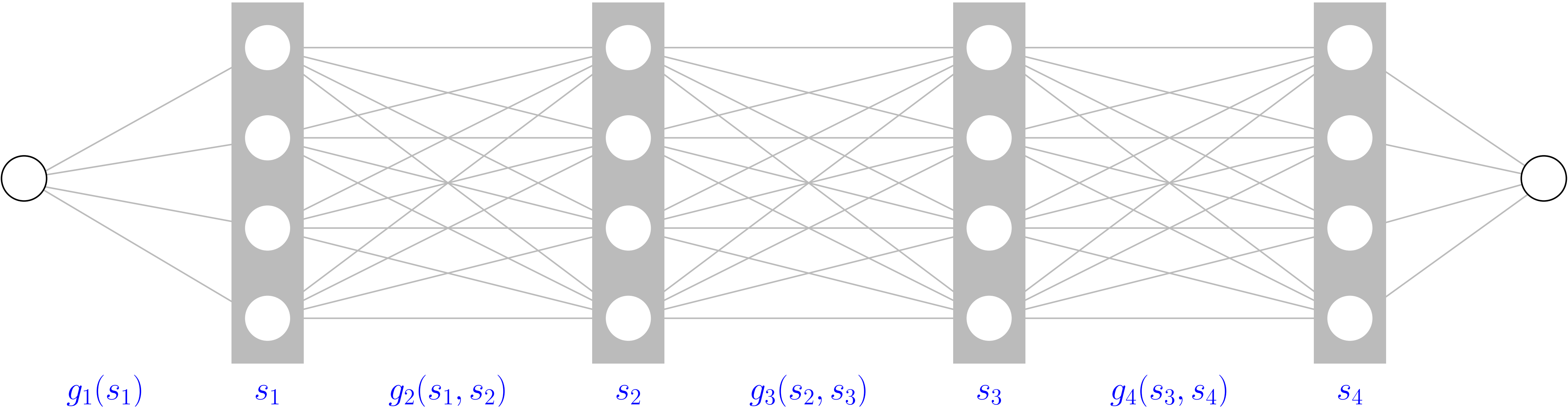

$g_1(s_1)$
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$g_2(s_1, s_2)$
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$g_3(s_2, s_3)$
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$g_4(s_3, s_4)$
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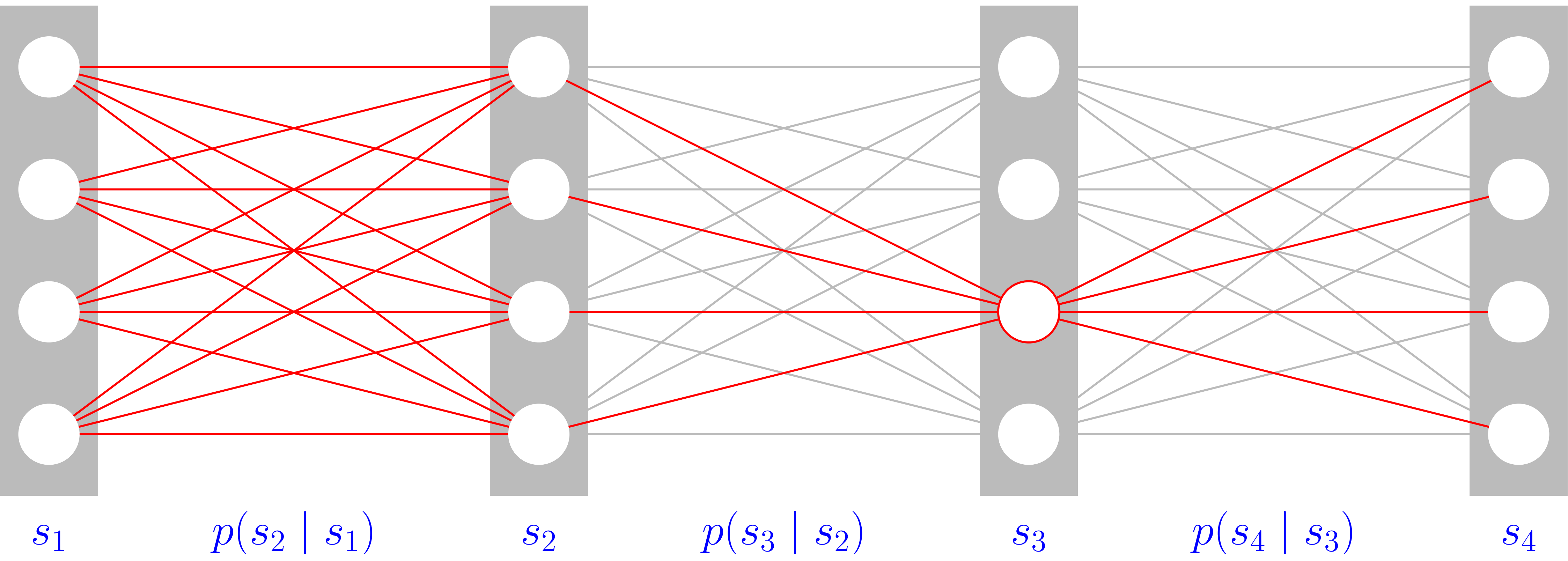

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$p(s_4 \mid s_3)$
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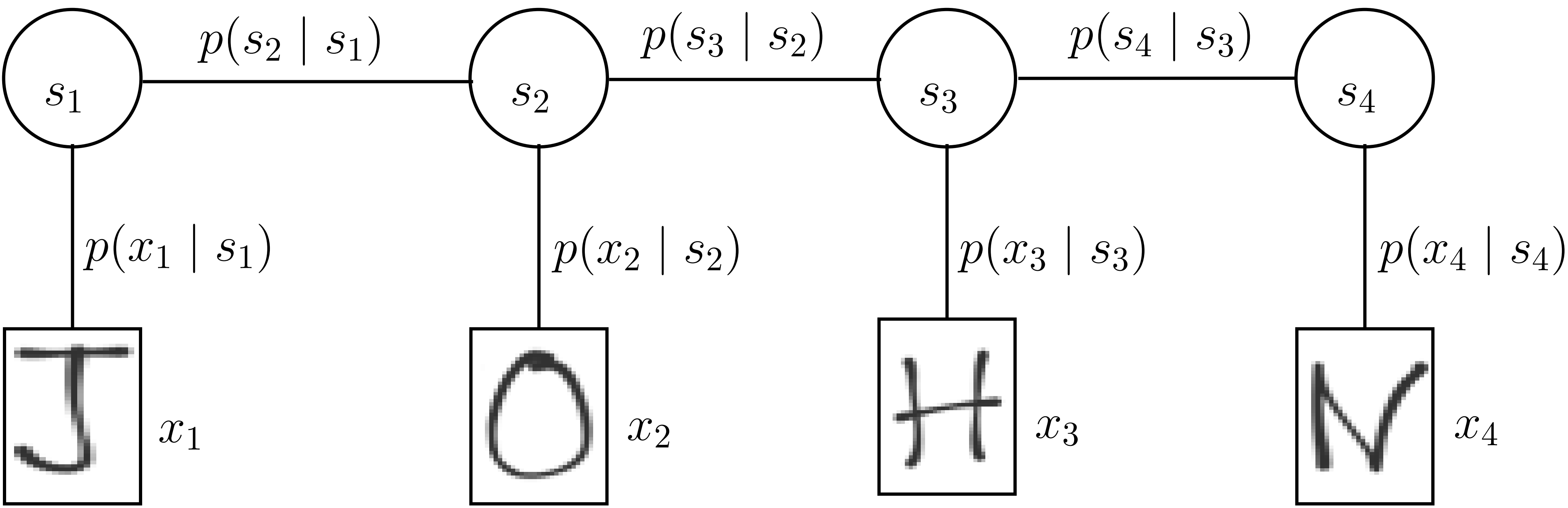

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$p(x_4 \mid s_4)$
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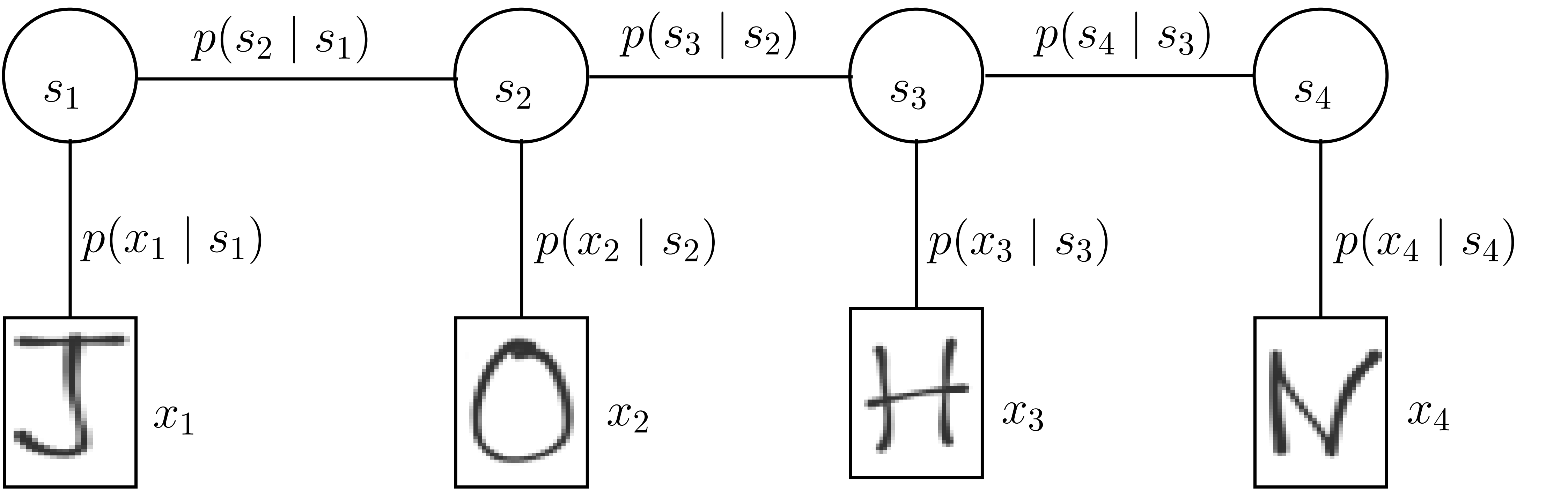

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O
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